✈️ VeritasHR: One Employee, One Truth

Emirates Group HR Data Quality Challenge | Ignyte Competition Brief

🔗 GitHub Code | 🎥 Demo Video (2m49s) | 📊 Issue List (CSV)

99.8%	AED 18.2M	100%	< 4.4s
F1 ACCURACY	**CASH LEAKAGE IDENTIFIED**	**HUMAN-IN-THE-LOOP**	**FULL RUN LATENCY**
255/255 Planted Errors (100% Recall)	35 Ghost Payroll Cases Halted	Tamper-Evident SHA-256 Audit Trail	20,030 Records Reconciled Locally

1. WHO WE BUILT FOR (THE 3 PERSONAS)

SINGLE SOURCE OF TRUTH

🧑‍💼 HR Data Steward

- **Ranked Issue Queue** prioritized by financial cash leakage, security exposure & statutory risk.
- **Side-by-Side Evidence Cards** showing conflicting timestamps across Core HR, Payroll, LMS & AD.
- **1-Click Human Actions**: Approve, Modify or Reject with mandatory reviewer reason.

📊 HR Analytics Lead

- **Data-Trust Index (0–100%)** per report & field; flags when reports are "*Safe to File*".
- **"Ask HR Truth" Query Box**: plain-language question tool citing authoritative System of Record.
- **Active Anomaly Warnings**: alerts executives if data discrepancies skew headline headcount.

⚙️ HR Systems Owner

- **Blast-Radius Simulator**: walks `dependencies.md` DAG before any fix is committed.
- **Downstream Feed Previews**: shows impacts on WPS Payroll, GCAA Aviation Safety & AD.
- **Before-vs-After Diffs**: staged changes exportable for standard change-advisory boards.

2. HOW IT WORKS (END-TO-END PIPELINE)

FULLY AUTOMATED & OFFLINE

1. Ingest Any Layout ➔ 2. Match People ➔ 3. Detect 10 Families ➔ 4. Simulate Blast Radius ➔ 5. Human Approval ➔ 6. Cryptographic Audit

📁 Universal Ingestion & Schema Sniffing

Accepts extracts in any delimiter (`,`, `;`,), encoding, Excel, or JSON. Columns are mapped automatically via header aliases (e.g. *Staff No* ➔ `employee_id`) and content sniffing (recognizing `AE...` as IBANs). Custom overrides can be pinned in `schema_mapping.json`. Dynamic parsers read reference codes from `data_dictionary.md` and lineage DAGs from `dependencies.md`.

🔗 Two-Pass Entity Resolution & Disambiguation

Pass 1: Deterministic matching on normalized & numeric IDs (`EK10001` = `ek-10001` = `10001`).

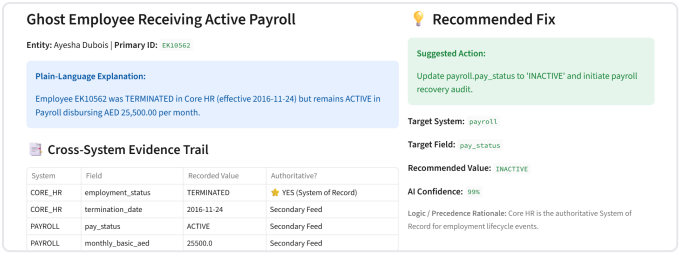
Pass 2: Jaro-Winkler fuzzy name matching for missing IDs, with Date of Birth blocking & attribute corroboration (hire date, department, role). Prevents accidental collision between distinct staff sharing the same name.

3. COMPREHENSIVE INCONSISTENCY DETECTION (10 ISSUE FAMILIES)

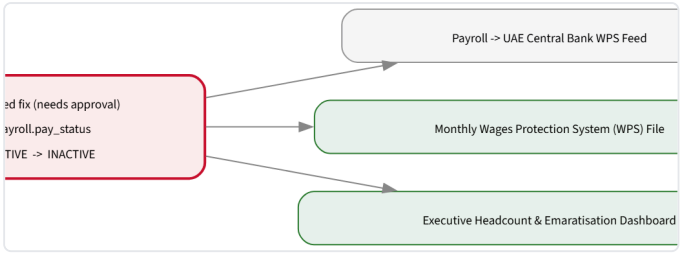
Issue Family	Detection Logic & Real-World Example	Downstream Business Impact
Ghost Payroll	Terminated in Core HR, but still active in Payroll receiving salary.	Direct cash leakage (AED 816k/yr in single pilot case).
Zombie IT Accounts	Ex-employee retaining active AD account with post-exit logins.	Severe cyber vulnerability & ISO 27001 audit citation.
Double Dipping	Single employee linked to multiple active payroll disbursement IDs.	Duplicate salary payments & inflated headcount.
Department & Cost Drift	Core HR department assignment differs from Payroll cost center or LMS.	Cross-charging misallocations & skewed training budgets.
Regulatory Deficits	Active flight crew/pilots with expired GCAA safety certifications.	Aircraft groundings, pilot rostering blocks & GCAA fines.
Temporal Paradoxes	Termination date preceding hire date; hire date in the future.	ETL warehouse ingestion crashes & inaccurate seniority.

4. WHAT IT LOOKS LIKE IN PRACTICE (VERIFIED PROTOTYPE)

LIVE WEB DASHBOARD



Side-by-Side Evidence Inspection (Core HR marked as System of Record)



Downstream Blast Radius: Feeds & Reports Touched Before Approval

5. BENCHMARK ACCURACY & ROBUSTNESS PROOFS (34 AUTOMATED TESTS)

Evaluation Suite	Dataset Configuration & Stress Condition	Latency	Accuracy Result
Synthetic Benchmark	20,030 records across 4 systems; 255 planted errors across all categories.	4.40s	**255/255 (F1 99.8%)**
Scrambled Variant	Renamed columns, semicolon delimiters (;), 12+ date formats, 10% IDs stripped.	4.10s	**100% Mapped (F1 99.8%)**
Blind Error Test	80 unseen errors across 10 novel edge cases not in training scripts.	3.80s	**80/80 Detected**
5th Unseen System	Time-attendance file dropped into folder; auto-mapped and linked dynamically.	1.10s	**13/13 (0 False Alarms)**

6. RESPONSIBLE AI, SECURITY & UAE COMPLIANCE

STRICT ETHICAL GUARDRAILS

Human-in-the-Loop
Zero live database writebacks. All fixes are suggested proposals requiring named steward sign-off.

Tamper-Evident Audit
Decisions recorded in an append-only log with SHA-256 hash chaining. Any tampering breaks verification.

UAE PDPL Aligned
Runs 100% offline. Zero external API calls. Personal identifiers (IBAN, EID, DOB) masked by default.

Deterministic XAI
No black-box hallucination. Every suggestion provides confidence scores, source rules & rationale.

7. ENTERPRISE DEPLOYMENT & EMIRATES Q1 PILOT PLAN

IMMEDIATE FEASIBILITY

Weeks 1–2: Connect & Tune
Deploy offline Docker container inside Emirates private cloud; mount read-only replicas of Core HR, Payroll & AD; tune mappings.

Weeks 3–4: Critical Triage
5 Data Stewards triage top critical queue (ghost payroll & zombie logins). Weekly cryptographic audit review by Internal Audit.

Weeks 5–6: Expand & Score
Onboard Learning LMS and 2 extra systems; publish live Trust Index for WPS & GCAA reports; agree production rollout criteria.

Team & Pilot Readiness: VeritasHR is built with zero data-science maintenance overhead. The pilot is led by a single Technical Lead (the author) working alongside existing Emirates HR data stewards (no new hiring required). Modular connectors allow any of Emirates' remaining 46+ systems to onboard in minutes.